IMVSC: An Improved Multiview Subspace Clustering in Multimodal Medical Image Application

Yanshuo Gong[1], Ru Nie[1,4](✉), Zhengwei Li[1,2,3], Lei Wang[1,2], and Xuanzhi Hu[1]

[1] School of Computer Science and Technology, China University of Mining and Technology, Xuzhou 221116, China
zwli@cumt.edu.cn
[2] Big Data and Intelligent Computing Research Center, Guangxi Academy of Science, Nanning 530007, China
[3] School of Information Science and Engineering, Zaozhuang University, Zaozhuang 277160, China
[4] Mine Digitization Engineering Research Center of the Ministry of Education, China University of Mining and Technology, Xuzhou 221116, China

Abstract. Multimodal medical image data provides a comprehensive view of patient conditions, and its effective utilization can enhance clinical decision-making. Traditional clustering methods face challenges in handling complex, complementary, and heterogeneous data. In this paper, we propose a novel method named IMVSC, which provides an integrative architecture for simultaneously optimizing the anchor learning to learn feature representations, graph construction to evaluate data similarity across views, and cluster partitioning to obtain the clustering results. Additionally, we incorporate adaptive variance selection based on the Bayesian Information Criterion, which can retain more informative features during the clustering process. Our IMVSC method exhibits outstanding clustering performance, achieving accuracy, normalized mutual information, purity, and F-score values of 97.09%, 88.38%, 97.16%, and 94.87%, respectively. These results highlight the potential of IMVSC in advancing medical decision-making and facilitating more precise analysis of multimodal medical image data.

Keywords: Multiview clustering · Anchor learning · Graph construction · Adaptive variance selection · Multimodal medical image

1 Introduction

Multimodal medical images integrate data from distinct imaging techniques, each offering distinct perspectives that, when appropriately utilized, can enhance clinical decision-making and support more accurate treatment planning [1, 2]. Different modalities, such as MRI, CT, and PET, provide complementary information: MRIs are particularly effective at capturing soft tissue details, while CT scans offer clearer images of bone structures.

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PET scans, on the other hand, are invaluable for assessing metabolic activity and detecting abnormal cellular functions, such as in cancer or neurological disorders. By effectively leveraging all modalities, clinicians can get a more comprehensive understanding of the patient's condition.

Multiview clustering methods are unsupervised techniques that cluster data samples by utilizing information from various views [3]. Multiview clustering methods aim to uncover underlying structure by utilizing complementary information from all views, thereby enhancing clustering quality [4, 5]. Each view represents a specific perspective of the multiview data [6]. When different views of multiview data are effectively leveraged, they can enhance the clustering performance [7, 8]. For multiview clustering analysis, two commonly employed methods are anchor le1arning and graph construction. Anchor learning selects representative anchor points and calculates the similarity between data points and anchors, extracting and integrating key information from multiple perspectives, thereby enabling clustering [9–12]. Graph construction builds similarity graphs for each perspective, and by optimizing the similarity matrices across multiple views, it achieves global representation of the data, ultimately achieving clustering [13–16]. However, anchor learning and graph construction each have its own strengths and limitations. Anchor learning is effective for cross-modal learning and high-dimensional data by selecting representative anchor points, but its random selection may miss global relationships. Graph construction, on the other hand, can effectively capture global relationships but struggles with high-dimensional data.

Recent advancements have significantly enhanced the application of multiview clustering methods in the medical field. Dornaika et al. proposed a solution for unsupervised clustering of chest X-ray images [17] with two constraints: a view-based smoothing constraint to ensure consistent cluster labeling across views, and a matrix orthogonality constraint to improve cluster delineation. Zhang et al. utilizes non-negative matrix factorization to predict multi-stage Alzheimer's disease (AD) progression by learning a consensus representation matrix [18]. Additionally, the Multiview Robust Graph-based Clustering (MRGC) method [19] tackles noise and high dimensionality in multi-omics data through robust representation learning, adaptive graph construction, and consensus structure learning.

Despite the advances in multiview clustering methods, research on multiview clustering analysis for multimodal medical images remains relatively underdeveloped. In existing studies [20–22], a common strategy is to first fuse different modalities of images and then apply clustering to fuse the results. However, this strategy generally assume that data from different modalities can be combined into a single feature space. Consequently, fusion methods might result in the loss of essential information. In contrast, multiview clustering methods address this issue by treating each modality separately. Instead of forcing the modalities into a single representation, multiview clustering preserves the unique relationships within each modality while also leveraging their complementary information. In this paper, to overcome the above shortcomings, we propose an improved multiview subspace clustering method for multimodal medical image applications (IMVSC).

2 Methodology

2.1 Proposed Multi-view Clustering Method

Self-representation in Clustering. This phase focuses on enhancing the self-representative matrix A to achieve two key objectives: enabling accurate data reconstruction through self-expression mechanisms and maintaining the dataset's essential topological characteristics.

Following the subspace clustering paradigm [23], the standard approach postulates linear representability between data instances, where the self-representation matrix quantifies these inter-sample relationships through linear combination coefficients.

$$\min_{A} \gamma \|X - D(X)A\|_q + \Omega(A), \tag{1}$$

where X is input data matrix, $D(X)$ is learnable dictionary matrix, and the initial value is generally X. A is also named affinity graph and stores the similarity between data points. Here, $\gamma > 0$ functions as a trade-off factor. The norm $| \cdot |_q$ indicates the square of the proper norm, and $\Omega(A)$ is the regularization term on A, refer to [29–31]. This equation aims to optimize A the representation of the original data by minimizing the reconstruction error (Fig. 1).

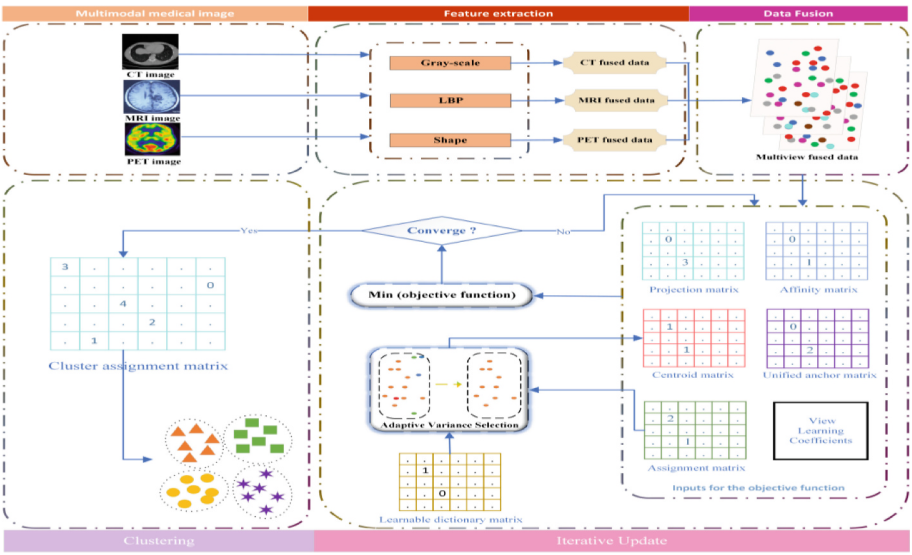

Fig. 1. Multiview clustering analysis procedure for a medical multimodal image.

From Singleview to Multiview. This stage extends self-representative dictionary learning to the multi-view scenario, aiming to learn the data representation for each view. The objective formulation for multiview subspace clustering extends the previous equation to minimize reconstruction errors across views as follows:

$$\min_{A} \gamma \sum_{v=1}^{m} \left\| X^{(v)} - X^{(v)} A^{(v)} \right\|_F^2 + \Omega(A^{(v)}), \tag{2}$$

where $A^{(v)}$ denotes the dictionary matrix for the v-th view, γ is a scalar weighting factor, and $\Omega\, A^{(v)}$ is a regularization term. The norm $| \cdot |_F$ represents the square of the Frobenius norm, reflecting the sum of the squares of a matrix's elements. Additionally, X_v denotes the v-th view of the dataset X, and constraints $A^{(v)} \geq 0$ ensuring nonnegativity, and $\sum_{j=1} A_{ij}^{(v)} = 1$, ensuring nonnegativity and the column sums of $A^{(v)}$ equaling 1.

Incorporating Anchor Learning. Building on the previous two steps of optimizing the self-representative dictionary matrix, this stage incorporate anchor learning to unify the representation of data from different views.

Mathematically, the optimization process aims to minimize the discrepancy between the affinity matrices across different views while ensuring the consistency of the anchor points. The optimization formula can be expressed as follows:

$$\min_{\beta, P^{(v)}, U, M} \sum_{v=1}^{m} \beta_V^2 \left\| X^{(vq)} - P^{(v)} U M \right\|_F^2 \tag{3}$$

where U is the unified anchor matrix, and its dimensions are $a \times b$, where a is the cross-view consistent latent dimension across different views and b is the anchor cardinality. X_v represents the v-th view of the original data, defined by its dimensions n and d_v, where n denotes the observation count and d_v indicates the view-specific feature dimensionality. α_v indicates the coefficients for each view, highlighting its role in the consensus affinity graph. M is the common affinity matrix, and its dimensions are $b \times n$. The projection matrix P_v aligns the unified anchor with each view's feature space of each view. β is the view coefficients, obtained through adaptive learning of the consensus affinity graph.

Specially, β, $P^{(v)}$, U, and M are model parameters subject to the following constraints: $B^T \mathbf{1} = 1$ ensures the normalization of weights, $P^{(v)T} P^{(v)} = I_d$ and $U^T U = I_l$ guarantee the orthogonality of the projection matrices and the matrix U, respectively. Non-negativity and sum normalization are imposed on M through $M \geq 0$ and $M^T \mathbf{1} = 1$.

Incorporating Graph Construction. The main objective of this optimization process is to unify multi-view data representation while accurately assigning each data point to its respective cluster.

$$\min_{\beta, P^{(v)}, U, M, Z, T} \sum_{v=1}^{m} \beta_V^2 \left\| X^{(vq)} - P^{(v)} U M \right\|_F^2 + \lambda \| M - ZT \|_F^2 \tag{4}$$

where Z represents the centroid matrix, and T denotes the cluster assignment matrix, $T_{ij} = 1$ indicating that the j-th instance is assigned to the i-th cluster, and 0 otherwise. Additionally, λ is the trade-off parameter.

Specially, the model parameters β, $P^{(v)}$, U, M, Z, and T adhere to several constraints. The normalization of the coefficients is assured by $B^T \mathbf{1} = 1$. Orthogonality in the projection matrices and the subspace basis matrices is maintained by $P^{(v)T} P^{(v)} = I_d$ and $U^T U$

$= I_l$, respectively. The subspace representation matrix M is subject to non-negativity and column-wise normalization conditions imposed by $M \geq 0$ and $M^T\mathbf{1} = 1$. The centroid matrix Z must be orthogonal, as specified by $Z^TZ = I_k$. Hard clustering constraints are defined by $T_{ij} \in \{0, 1\}$ and $\sum_{i=1}^{k} T_{ij} = 1$ for all j, which ensures each instance is assigned to exactly one cluster.

Iterative Update. However, there is still an optimization problem for variables in (4). Each subproblem is a convex optimization problem [29], allowing for the optimization of each variable individually while keeping the others fixed.

Algorithm1 Iterative Framework for Variable Optimization

Input: Multiview dataset, trade-off parameter $\lambda > 0$, and cluster number k.

Initialize: $P^{(v)} = U = M = 0$, $Z = T = I$.

while not converged **do**

Update $P^{(v)}$, $\forall v$ by solving the problem in (5).

Update U by solving the problem in (6).

Update M by solving the problem in (7).

Update Z by solving the problem in (8).

Update T by solving the problem in (9).

Update β_v by solving the problem in (10).

end while

Output: Final consensus cluster assignment matrix T.

The objective function of $P^{(v)}$ can be reformulated as

$$\min_{P^{(v)}} \sum_{v=1}^{m} \beta_V^2 \left\| X^{(v)} - P^{(v)}UM \right\|_F^2 \tag{5}$$

and the orthogonality of $P^{(v)}$ is ensured by the constraint $P^{(v)T}P^{(v)} = I_a$, where I_a is the a $\times$ a identity matrix.

The objective function of U can be reformulated as

$$\min_{U} \sum_{v=1}^{m} \beta_V^2 \left\| X^{(vq)} - P^{(v)}UM \right\|_F^2, \quad s.t.\ U^TU = I_b \tag{6}$$

where I_b is the identity matrix of size b.

The objective function of M can be reformulated as

$$\min_{M} \sum_{v=1}^{m} \beta_V^2 \left\| X^{(v)} - P^{(v)}UM \right\|_F^2, + \lambda \| M - ZT \|_F^2 \tag{7}$$

and the constraints ensure that each $M_{:j}$ is normalized and non-negative, specified by $M_{:j}^T\mathbf{1} = 1$ and $M \geq 0$.

The objective function of Z can be reformulated as

$$\min_{Z} \lambda |M - ZT|_F^2, \quad s.t.\ Z^T Z = I_K \tag{8}$$

The objective function of T can be reformulated as

$$\min_{Z} \lambda |M - ZT|_F^2, \quad s.t.\ T_{ij} \in \{0, 1\}, \ \sum_{i=1}^{k} T_{ij} = 1, \quad \forall j = 1, 2, \ldots, n \tag{9}$$

The objective function of β_v can be reformulated as

$$\min_{\beta} \sum_{v=1}^{m} \beta_v^2 R^{(v)^2}, \quad s.t.\ \beta^T \mathbf{1} = 1 \tag{10}$$

2.2 Adaptive Variance Selection

To improve clustering performance, we designed adaptive variance selection, ensuring that the most informative features are retained.

Algorithm2 Adaptive Variance Selection

Input: Learnable dictionary matrix $A^{(v)}$ and the cluster assignment matrix T, target explained variance threshold ξ, minimum number of classes $numclass$.

Output: Dynamically optimized centroid matrix Z.

Step 1: Concatenate matrices A and T to form J. See (11).

Step 2: Perform SVD on J, calculate cumulative explained variance, and determine the dimensionality k for a variance threshold ξ. Refer to (12) and (13).

Step 3: Compute BIC for each k and select the optimal k that minimizes the BIC, ensuring $k \geq numclass$. See (14) and (15).

Step 4: Construct centroid Z using the first k singular vectors corresponding to the largest singular values. These singular vectors represent the most informative features in the data.

$$J = A \cdot T^\top \tag{11}$$

where J is the resulting matrix formed by the horizontal concatenation of matrices A and T, with $T^\top$ being the transpose of matrix T. This matrix J is SVD.

$$J = U \Sigma V^\top \tag{12}$$

where J is the matrix derived from the singular value decomposition (SVD) of a given matrix, with U and V as the left and right singular vector matrices, respectively, and $\sum$ being the diagonal matrix of singular values.

$$CEV_k = \frac{\sum_{i=1}^{k} \sigma_i}{\sum_{i=1}^{n} \sigma_i} \tag{13}$$

where $\mathrm{CEV_k}$ represents the cumulative explained variance, computed to determine how many dimensions k are required to capture at least ξ percent of the total variance. Here, σ_i represents the individual singular values, and n denotes the total number of singular values, which corresponds to the dimensionality of the data matrix. Specially, the value of ξ is generally taken to be around four-fifths.

$$BIC(k) = n \ \log\left(\frac{RSS(k)}{n}\right) + k \ \log(n) \tag{14}$$

where the *BIC(k)* is determined for each value of k to calculate the optimal dimensionality by balancing the model fit and complexity, with *RSS(k)*being the residual sum of squares for k dimensions, and n the number of observations.

$$best\_k = \max(best\_k, numclass) \tag{15}$$

where this step ensures that the selected k is large enough to represent all distinct classes, with *numclass*. Specifying the cardinality of distinct categories.

3 Experiments

3.1 Datasets and Compared Algorithms

We assessed the clustering performance of the IMVSC method by comparing it with six widely used clustering algorithms. The assessment employed five widely recognized multimodal medical image datasets.

Datasets. The following datasets were utilized for evaluation:

CHAOS: Comprising multi-organ segmentation data from CT and MRI modalities, this dataset includes labels for the liver, kidney, and spleen. **TCGA-DD-A1EH04**: Featuring modalities such as SCOUT scans, biphasic CT scans, high-resolution thin-layer scans, CT coronal plane images, and maximum intensity projection images. **TCGA-DD-A1EI03**: Comprising modalities like SCOUT scans, biphasic CT scans, and CT coronal plane images. **TCGA-DD-A1EJ06**: Includes SCOUT scans, abdominal CT scans, dual contrast CT scans, and reformatted coronal plane images. **MMWHS**: A comprehensive dataset primarily labeled for cardiac segmentation in both CT and MRI modalities, focused on the left atrium.

Compared Algorithms. We evaluate the proposed method against several established and efficient baselines widely recognized in the field: **K-Means**: A well-known clustering method [24]. **Hierarchical Clustering**: This method builds a dendrogram by progressively merging similar groups [25]. **Quality Threshold Clustering (QTC)**: A density-based algorithm [26]. **The Gaussian Mixture Model (GMM)** is a probabilistic framework that posits data points originate from a combination of Gaussian distributions [27]. **Multiview Spectral Clustering (MVSC)**: A traditional method that integrates data from multiple viewpoints to unearth underlying structures [28]. **Efficient Orthogonal Multiview Subspace Clustering (OMSC)**: Tailored for large-scale data [29]. Each algorithm was assessed using the optimal parameters and settings recommended in the literature [24–29].

3.2 Results

Table 1 illustrates that our IMVSC method significantly outperforms traditional clustering algorithms across multiple datasets. Specifically, for the CHAOS dataset, IMVSC achieved the highest scores in all metrics. Although the Fscore and NMI were lower,

Table 1. Comparison of the effectiveness of IMVSC with other clustering algorithms.

Datasets	Metric	K-Means	Hierarchical	QTC	GMM	MVSC	OMSC	IMVSC
CHAOS	ACC	0.92483	0.66559	0.44033	0.93640	0.81799	0.92091	**0.97086**
	NMI	0.59783	0.44236	0.07948	0.64265	0.52362	0.66885	**0.82156**
	Purity	0.92483	0.92830	0.76642	0.93640	0.92946	0.92068	**0.97155**
	Fscore	0.86975	0.65239	0.50081	0.88846	0.82662	0.85879	**0.94866**
TCGA-DD-A1EH04	ACC	0.45406	0.50394	0.37008	0.55906	0.50919	0.62205	**0.64567**
	NMI	0.13897	0.17019	0.10740	0.16474	0.15978	0.41647	**0.42075**
	Purity	0.55643	0.60236	0.54331	0.61286	0.51181	0.72178	**0.77428**
	Fscore	0.40130	0.44844	0.44543	0.50161	0.57752	0.60632	**0.63483**
TCGA-DD-A1EI03	ACC	0.70877	0.56491	0.60351	0.71228	0.61053	0.78596	**0.94035**
	NMI	0.08216	0.11173	0.07036	0.33595	0.13183	0.69294	**0.88376**
	Purity	0.76057	0.74386	0.70175	0.85965	0.76140	0.91579	**0.94035**
	Fscore	0.66680	0.57021	0.64311	0.68749	0.61470	0.77693	**0.91002**
TCGA-DD-A1EJ06	ACC	0.39662	0.45991	0.27426	0.48523	0.45992	0.60759	**0.62447**
	NMI	0.20003	0.18451	0.15159	0.21038	0.19663	0.47543	**0.47345**
	Purity	0.49367	0.46835	0.42616	0.53164	0.46835	0.67511	**0.69198**
	Fscore	0.35903	0.39138	0.35447	0.40892	0.43542	0.55452	**0.58094**
MMWHS	ACC	0.82476	0.66002	0.50053	0.74711	0.81427	0.71354	**0.85414**
	NMI	0.33437	0.34231	0.10535	0.49917	0.40513	0.51313	**0.47236**
	Purity	0.82476	0.82791	0.68625	0.86568	0.83106	0.85414	**0.85939**
	Fscore	0.73072	0.62471	0.59052	0.67537	0.74950	0.67485	**0.77503**

they still represented an improvement over baseline methods, including K-Means and Hierarchical Clustering.

3.3 Parameter Analysis and Convergence Analysis

This subsection presents a parameter analysis of the trade-off parameter λ across various benchmark datasets. Figure 2 illustrates the impact of parameter values on ACC and NMI across all benchmark datasets. The clustering performance of IMVSC remains consistently stable across different parameter value ranges on various datasets. Consequently, the proposed method demonstrates robustness concerning the parameter λ. The figure also provides the best results across various datasets. Optimal performance on the CHAOS dataset is achieved when λ is 0.01. Similarly, the figure provides the optimal results for the other datasets.

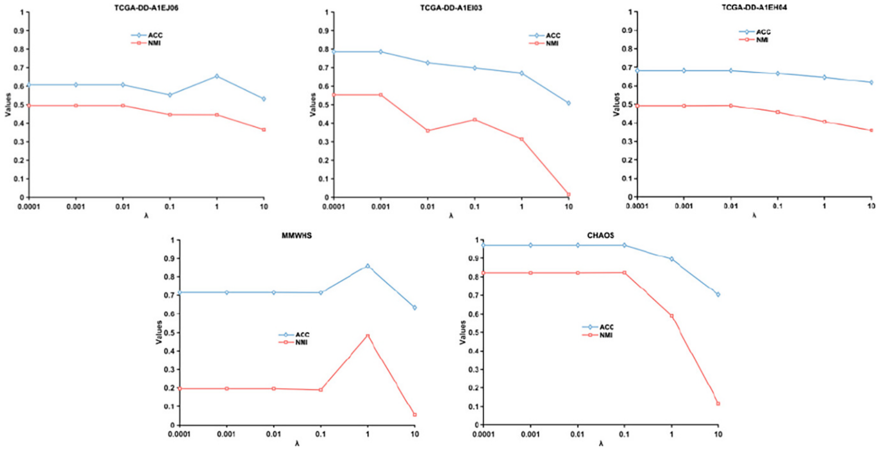

Fig. 2. The impact of varying trade-off parameters λ on performance.

In this subsection, we record the values of the objective function in (4) across various benchmark datasets to verify the convergence properties of the proposed method. Figure 3 illustrates the progression of these objective values across all benchmark datasets. The objective value decreases quickly during the initial two to five iterations and then stabilizes with additional iterations. Notably, for certain datasets, the objective value attains its minimum after just two or three iterations, thereby demonstrating the rapid convergence of our proposed IMVSC method.

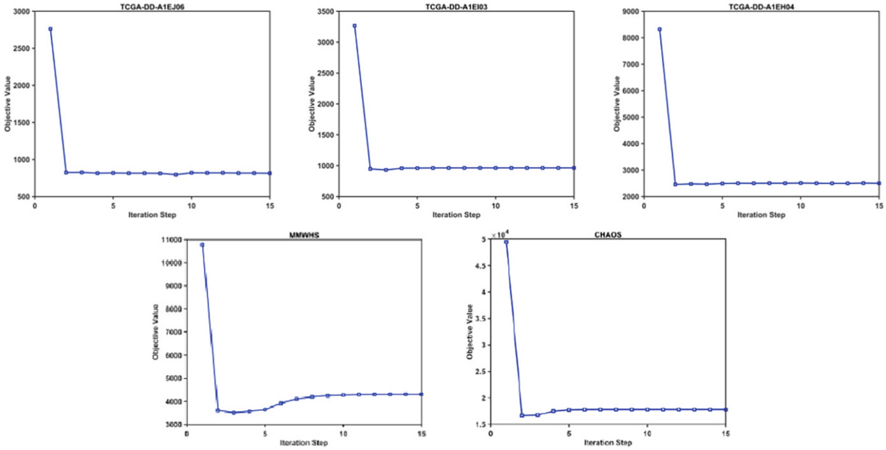

Fig. 3. The objective function value variation with iteration step.

4 Conclusion

In this paper, we propose a framework named IMVSC for clustering multimodal medical image data, and experiments attested to its efficacy and reliability. Furthermore, to improve clustering performance, this study incorporates adaptive variance selection, the effectiveness of which has been demonstrated through ablation experiments. Unlike previous research, our framework is unified and designed for the joint optimization of anchor learning to identify feature representations, graph construction to assess data similarity across views, and cluster partitioning.

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Disclosure of Interests. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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